COURSE NAME

DISCRETE MATHEMATICS

CSC 1204

(UNDERGRADUATE)

CHAPTER 4

FUNCTIONS

M. MAHMUDUL HASAN

ASSISTANT PROFESSOR, CS, AIUB

http://www.dit.hua.gr/~m.hasan







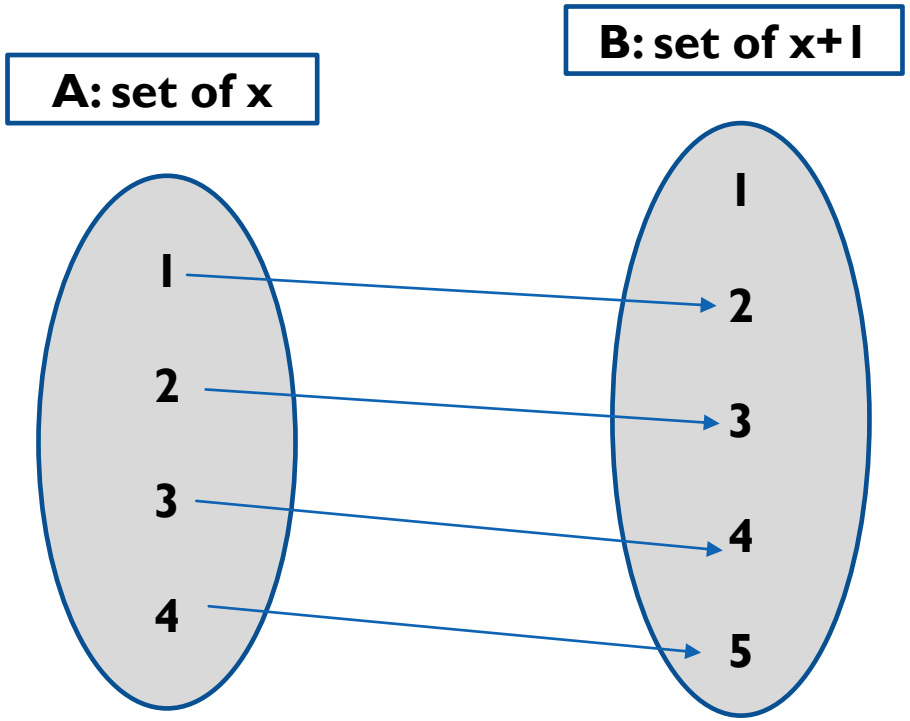
FUNCTIONS

- Let A and B be nonempty sets.

  A *function f* from *A* to *B* is an assignment of exactly one element of *B* to each element of *A*.

  We write $f(a) = b$ if *b* is the unique element of B assigned by the function *f* to the element *a* of A.

- If *f* is a function from A to B, we write $f: A \rightarrow B$
- Functions are also called *mappings* or *transformations*.
- Often we give a formula, such as f(x) = x + 1, to define a function.



TERMINOLOGY IN FUNCTIONS

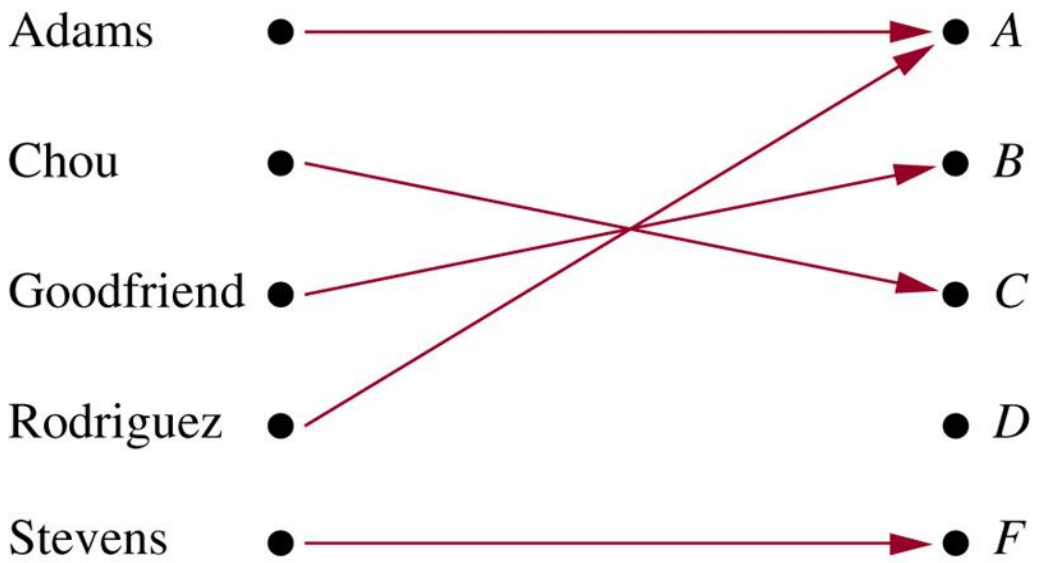
- *If f is a function from A to B, we say that f maps from A to B*

  $f: A \rightarrow B$

  - *A: domain of f*
  - *B: codomain of f*
  - *f(a)=b,*
    - *a is called the preimage of b*
    - *b is called the image of a*
  - *Range of f :* the set of all images of elements of *A* which is {*A, B, C, F*}

FIGURE 1: ASSIGNMENT OF GRADES IN A DISCRETE MATHEMATICS CLASS

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- Suppose *f* is a function mapping students in this class to the set of grades {*A, B, C,D,F*}
- *f*'s codomain is: {*A, B, C,D,F*}, and it's range is unknown!
- Suppose the grades turn out all *A*s and *B*s.
- Then the range of *f* is {*A, B*}", but it's codomain is still {*A, B, C,D,F*}.

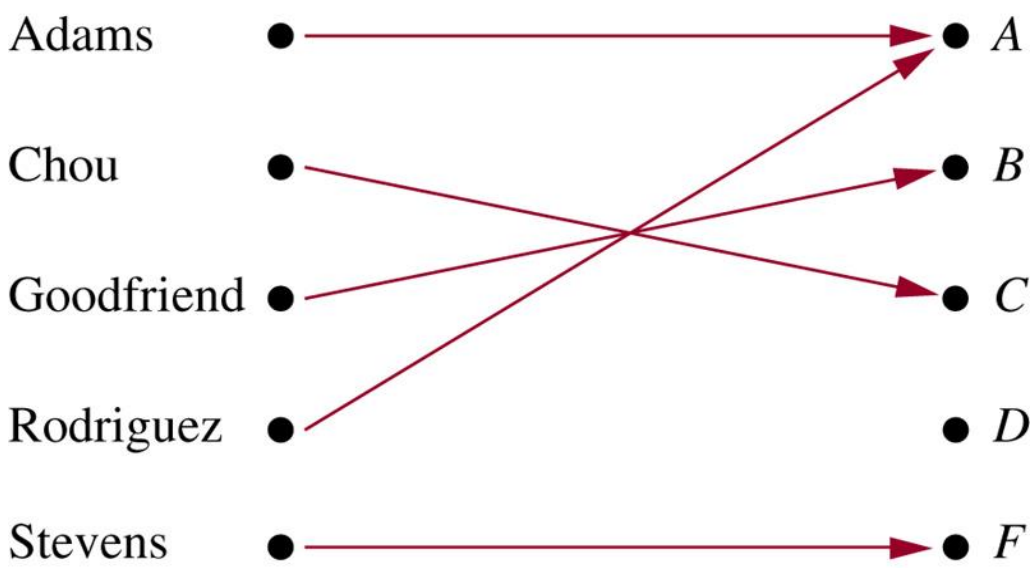
RANGE VS. CODOMAIN

Q: What are the **domain**, **codomain**, and **range** of the function that assigns grades to students of Discrete Math class as follows?

Solution:

- Let G be the function that assigns grade to a student of DM class.
- The **domain of G** is the set { Adams, Chou, Goodfriend, Rodriguez, Stevens}
- The **codomain of G** is the set { A, B, C, D, F}
- The **range of G** is the set { A, B, C, F} Because each grade except D is assigned to some student

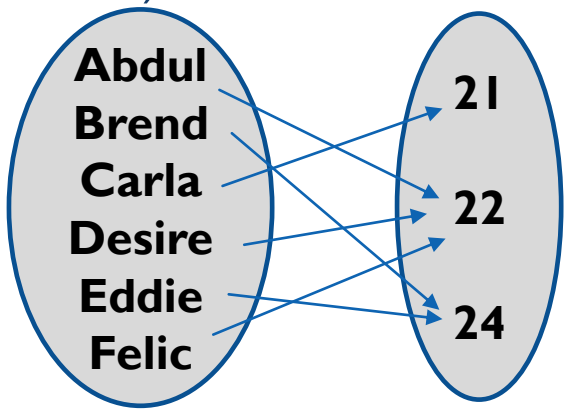
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RANGE VS. CODOMAIN

Q: Let R be the relation consisting of ordered pairs (Abdul, 22), (Brenda,24), (Carla,21), (Desire,22), (Eddie,24), and (Felicia,22), where each pair consists of a graduate student and the age of this student. *What is the function that this relation determines?*

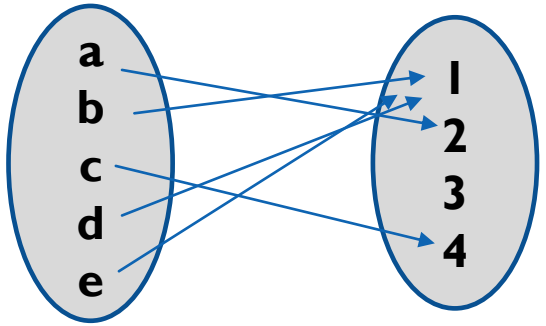
Solution: This relation defines the function f, where with f(Abdul)= 22, f(Brenda)=24, f(Carla)=21, f(Desire)=22, f(Eddie)= 24, and f(Felicia)=22.



- Here, **domain** is the set {Abdul, Brenda, Carla, Desire, Eddie, Felicia}
- **codomain** to be the set of positive integers {21, 22, 24}
- **Range** is the set {21,22,24}

Q: Let A = {a, b, c, d, e} and B = {1, 2, 3, 4} with f(a)=2, f(b)=1, f(c)= 4, f(d)=1, f(e)=1. What is the image of the subset S = {b, c, d}?

Solution: The image of the subset S = {b, c, d} is the set f(S) = {1, 4}



FUNCTIONS

- Let f_1 and f_2 be functions from A to **R**. Then f_1+f_2 and f_1f_2 are also functions from A to **R** defined by
  - $(f_1+f_2)(x) = f_1(x)+f_2(x)$
  - $(f_1f_2)(x) = f_1(x)f_2(x)$

Q. Let f_1 and f_2 be functions from **R** to **R** such that $f_1(x)=x^2$ and $f_2(x)=x - x^2$. What are the functions f_1+f_2 and f_1f_2 ?

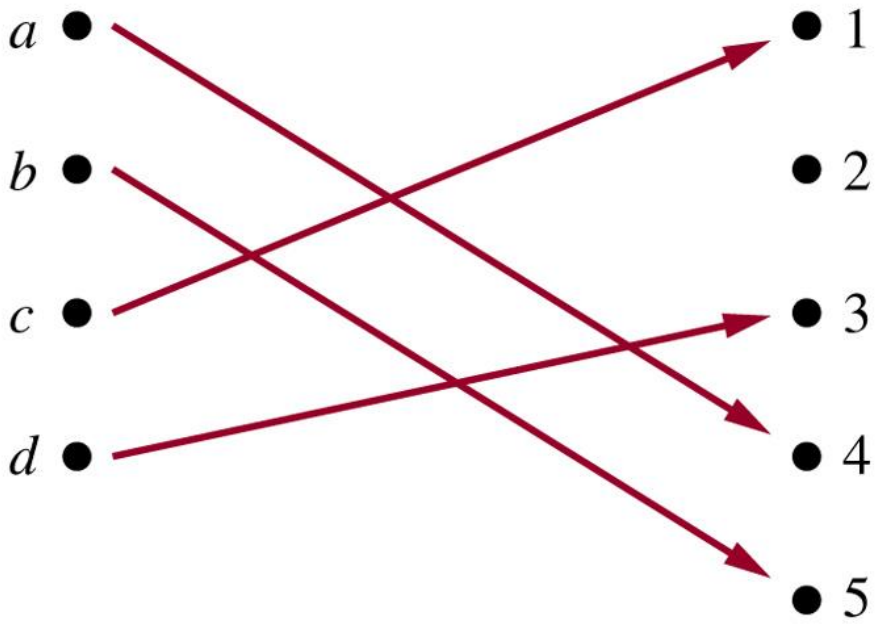
Solution:

- $(f_1+f_2)(x) = f_1(x)+f_2(x) = x^2 + (x - x^2) = x$
- $(f_1f_2)(x) = x^2(x- x^2) = x^3 - x^4$

ONE-TO-ONE FUNCTIONS

- A function f is *one-to-one* or *injective*, if $f(a)=f(b)$ implies that $a=b$ for all a and b in the domain of f.
  - $\forall a \forall b(f(a)=f(b) \rightarrow a=b)$ *or*
    $\forall a \forall b(a \neq b \rightarrow f(a) \neq f(b))$
- A function $f: A \rightarrow B$ is said to be one-to-one if different elements in the domain A have **distinct** images.
- A function is one-to-one, if every element of its range has exactly one pre-image.

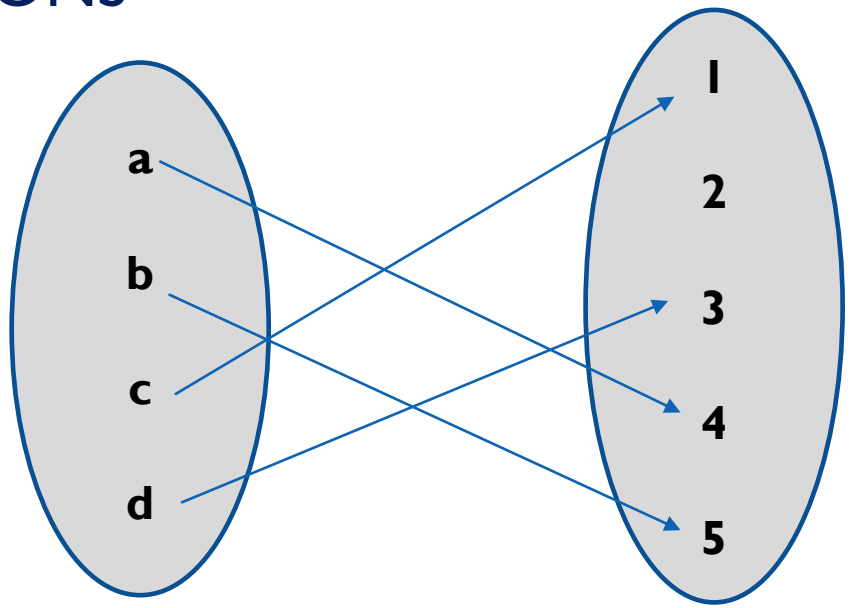
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ONE-TO-ONE FUNCTIONS

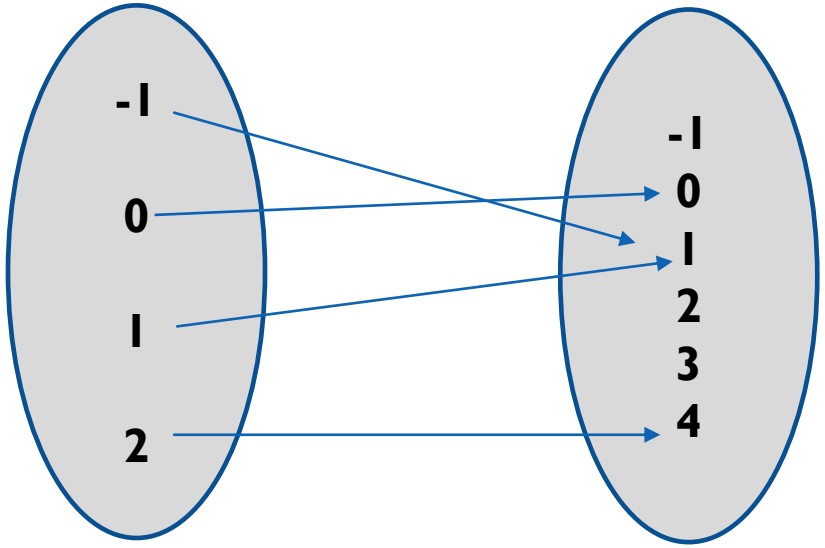
Q. Determine whether the function f from {a, b, c, d} to {1, 2, 3, 4, 5} with f(a)=4, f(b)=5, f(c)= 1, and f(d)=3 is one-to-one.

Solution: The function *f* is one-to-one because *f* takes on different values at the four elements of its domain.



Q. Determine whether the function $f(x) = x^2$ from the set of integers (Z) to the set of integers (Z) is one-to-one.

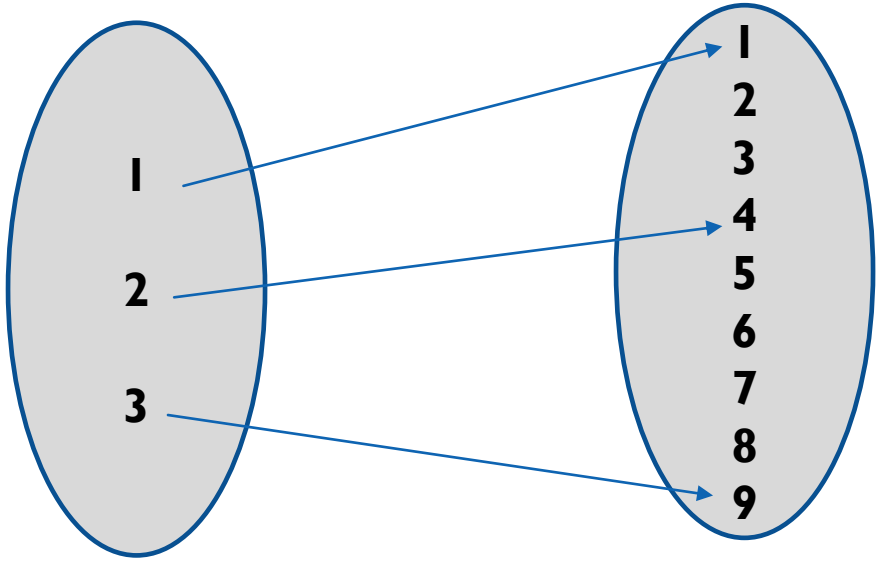
Solution: The function $f(x) = x^2$ is ***not*** one-to-one because, for instance, f(1)= 1 also f(–1) = 1 which implies the a domain elements can have multiple image.



ONE-TO-ONE FUNCTIONS

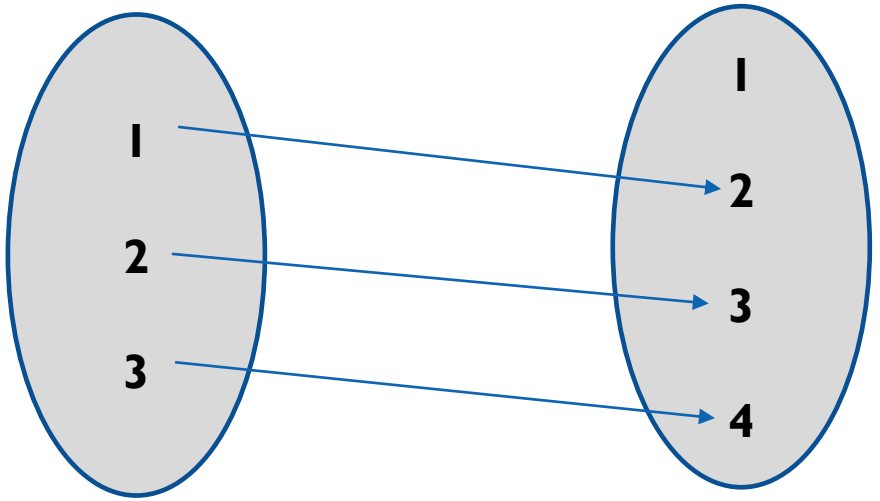
Q. Determine whether the function $f(x) = x^2$ from the set of positive integers to the set of positive integers is one-to-one.

Solution: The function *f* is one-to-one because *f* takes on different values at all the elements of its domain.



Q. Determine whether the function $f(x) = x + 1$ from the set of real numbers to the set of real numbers is one-to-one.

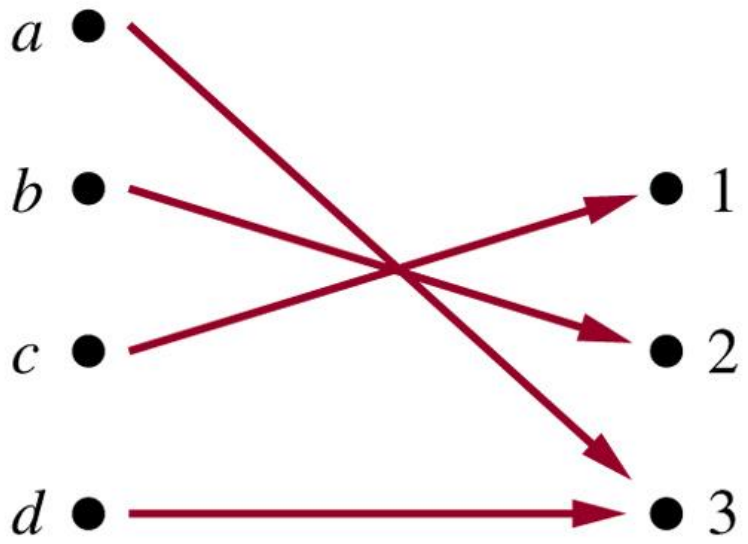
Solution: The function $f(x) = x + 1$ is a one-to-one function. Since $x+1 \neq y+1$, when $x \neq y$



ONTO FUNCTIONS

- A function f from A to B is ***onto*** or ***surjective***, if for every element $b \in B$ there is an element $a \in A$ with $f(a)=b$.
  - $\forall y \exists x(f(x)=y)$ **or** $\forall a \forall b(a \neq b \rightarrow f(a) \neq f(b))$
- A function $f: A \rightarrow B$ is said to be an ***onto*** function if each element of B is the image of ***some*** element of A
- A function f from A to B is onto if every element of B is the image of ***some*** element in A, i.e., if B = range of f

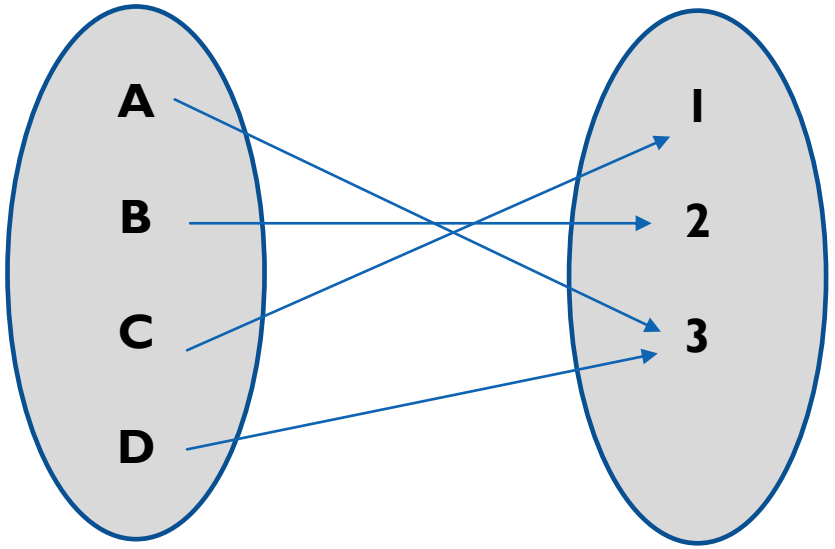
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ONTO FUNCTIONS

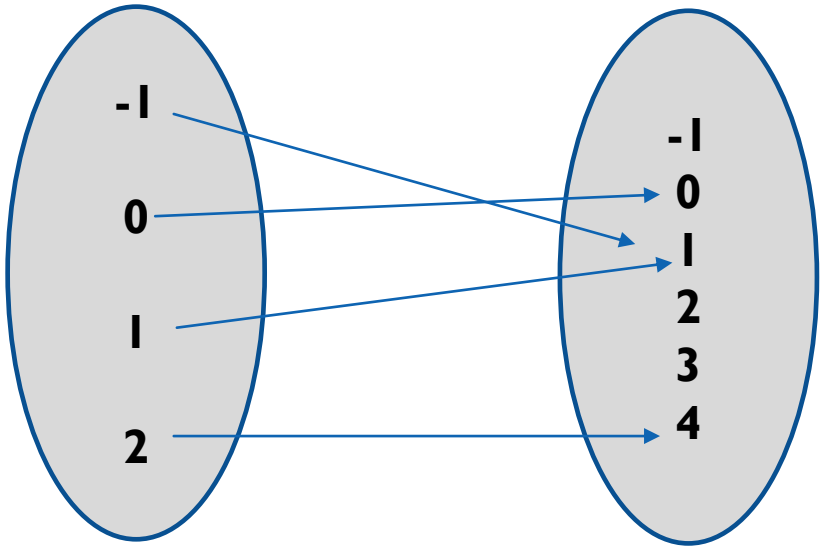
Q. Let f be the function from {a, b, c, d} to {1, 2, 3} defined by f(a)=3, f(b)=2, f(c)=1, and f(d)=3.
Is the function f an onto?

Solution: Because all three elements of the codomain are images of elements in the domain, therefore, *f* is onto.



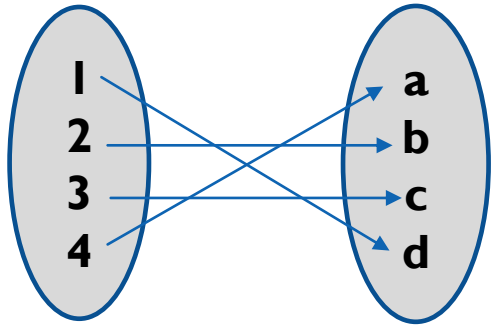
Q. Is the function $f(x)=x^2$ from the set of integers to the set of integers onto? f: Z → Z, here Z is integer number, Z+ is positive integer number

Solution: The function *f* is not onto, because there is no integer x with $x^2 = -1$, for instance.

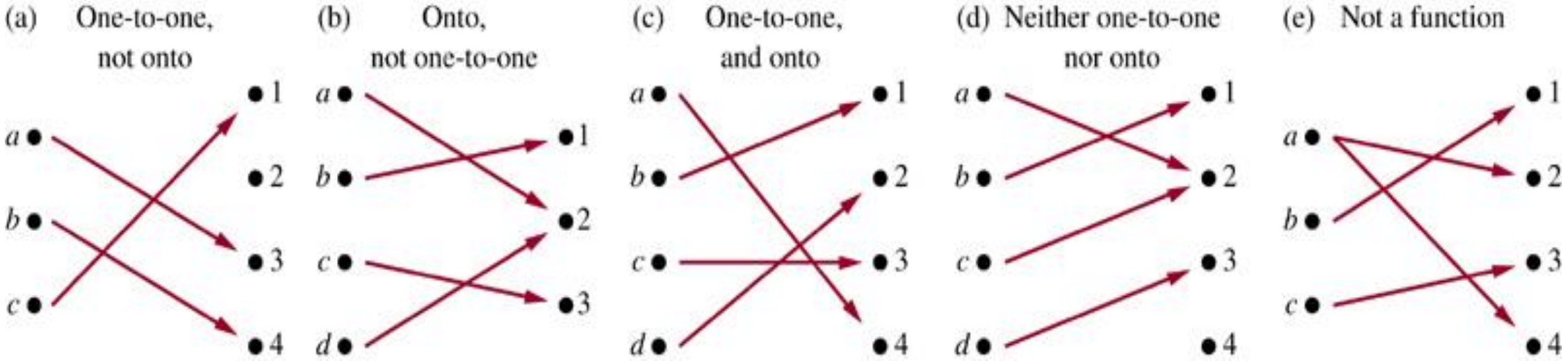


ONE-TO-ONE CORRESPONDENCE (*BIJECTION*)

- A function *f* is a ***one-to-one correspondence*** or a ***bijection*** if it is both one-to-one and onto.
- **Example**: Let ***f*** be the function from A to B where A={1, 2, 3, 4} and B = {a, b, c, d} with *f*(1)=d, *f*(2)=b, *f*(3)=c, and *f*(4)=a, then *f* is bijective function.
  - *f* is one-to-one since the function takes on distinct values.
  - *f* is onto since every element of B is the image of some element in A.



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INVERSE FUNCTIONS

- Let f be a one-to-one correspondence from A to B. The inverse function of f is the function that assigns to an element b belonging to B the unique element a in A such that f(a)=b. Hence,

  $f^{-1}(b) = a$, when $f(a) = b$

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- A one-to-one correspondence is called **invertible** because we can define an inverse of this function.
- A function is not invertible if it is not a one-to-one correspondence, because the inverse of such a function does not exist.

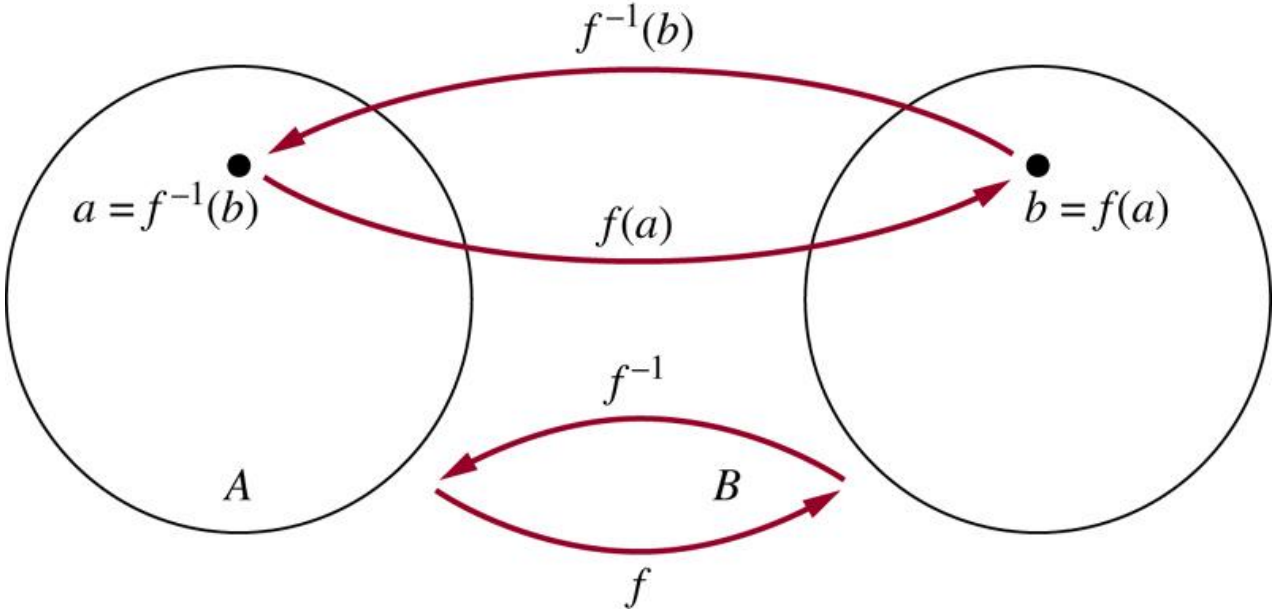


FIGURE THE FUNCTION F^{-1} IS THE INVERSE OF FUNCTION F

INVERSE FUNCTIONS

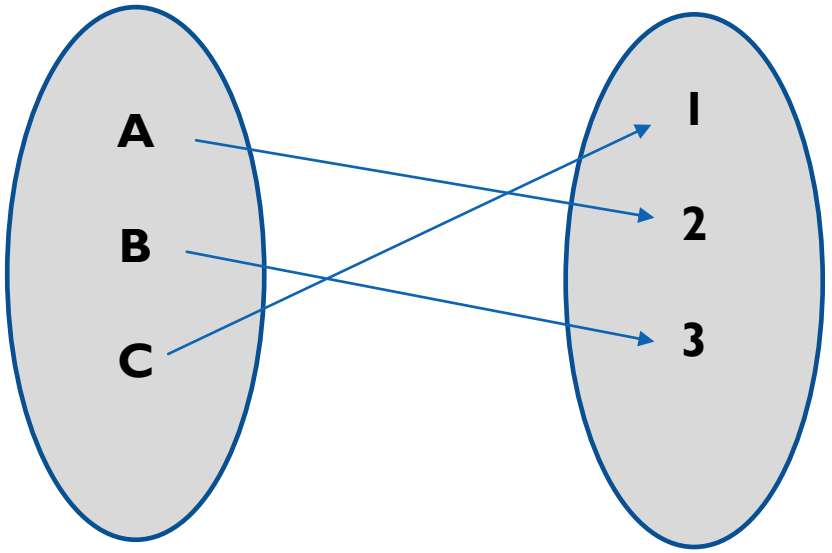
Q. Let f be the function from {a, b, c} to {1, 2, 3} such that f(a)=2, f(b)=3, and f(c)=1. Is f invertible, and if it is, what is its inverse?

Solution: The function f is invertible because it is a one-to-one correspondence. The inverse function f^{-1} reverses the correspondence given by *f*. So,

$f^{-1}(1)=c$,

$f^{-1}(2)=a$,

$f^{-1}(3)=b$



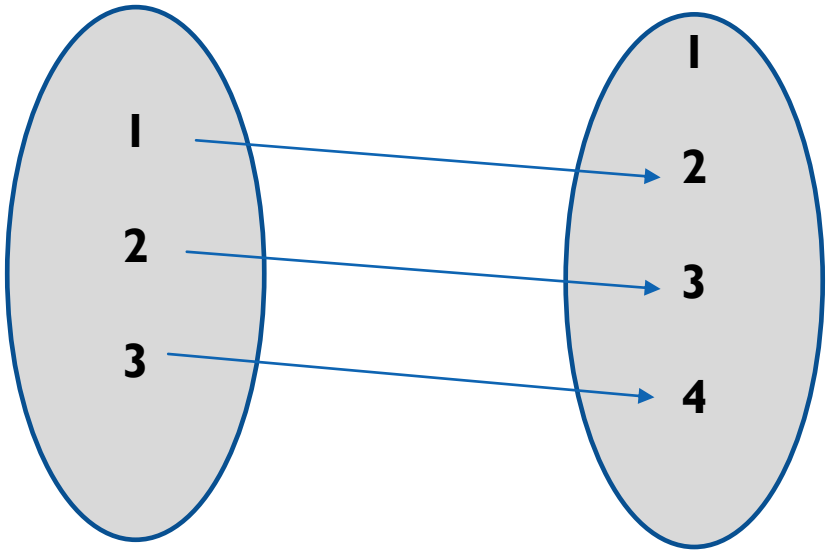
INVERSE FUNCTIONS

Q. Let $f: Z \rightarrow Z$ be such that f(x) = x +1. Is f invertible? and if it is, what is its inverse?

Solution: The function has an inverse because it is a one-to-one correspondence, as we have shown.

To reverse the correspondence, suppose that y is the image of x, so that y = x+1. Then x = y – 1.

This means that y – 1 is the unique element of Z that is sent to y by f. Consequently, $f^{-1}(y) = y - 1$



Let f(x) = x+1 = y
So, x = y-1

f(x) = y,
$f^{-1}(y) = x$
$f^{-1}(y) = y-1$

COMPOSITION OF FUNCTIONS

- Let g be a function from the set ***A*** to the set ***B*** and let f be a function from the set ***B*** to the set ***C***. The compositions of functions f and g, denoted by $f \circ g$, is defined by

  $(f \circ g)(a) = f(g(a))$

  Note: $f \circ g$ and $g \circ f$ are not equal

  Note: The composition $f \circ g$ can NOT be defined ***unless*** the range of g is a subset of the domain of f

Q. Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. Find the composite functions of f and g.

Solution:

- The composition of the functions f and g is

  $(f \circ g)(x) = f(g(x)) = f(3x+2) = 2(3x+2) + 3 = 6x + 7$

- The composition of the functions g and f *is*

  $(g \circ f)(x) = g(f(x)) = g(2x+3) = 3(2x+3) + 2 = 6x + 11$

Find $f \circ g$ and $g \circ f$ when x=2

$(f \circ g)(x) = 6x + 7$

$= 6x2 + 7$

$= 19$

$(g \circ f)(x) = 6x + 11 = 23$

FLOOR & CEILING FUNCTIONS

- The **floor** and **ceiling** functions map the real numbers onto the integers (**R**→**Z**).

Floor function

- Assigns to the real number x the largest integer that is less than or equal to x.
- Let x be a real number. The floor function rounds x down to the closest integer less than or equal to x.
- The value of the floor function at x is denoted by $\lfloor x \rfloor$
- **Examples:** $\lfloor 2.3 \rfloor = 2, \lfloor 2 \rfloor = 2, \lfloor 0.5 \rfloor = 0, \lfloor -3.5 \rfloor = -4$

Ceiling function

- ceiling function assigns to the real number x the smallest integer that is greater than or equal to x.
- The value of the ceiling function at x is denoted by $\lceil x \rceil$
- **Examples:** $\lceil 2.3 \rceil = 3, \lceil 2 \rceil = 2, \lceil 0.5 \rceil = 1, \lceil -3.5 \rceil = -3$

FLOOR & CEILING FUNCTIONS

Q. Find out the values using floor and celling function of $\lfloor \lceil 7/4 \rceil + 2 - \lfloor 13/3 \rfloor \rfloor$

$$\lfloor \lceil 7/4 \rceil + 2 - \lfloor 13/3 \rfloor \rfloor$$
$$= \lfloor 2 + 2 - 4 \rfloor$$
$$= \lfloor 0 \rfloor$$
$$= 0$$

Q. How many **bytes** are required to encode 600 kilobits of data? [***Note***: Each byte is made up of 8 bits]

Answer: $\lceil (600X1000)/8 \rceil$ = 75000 bytes

Q. How many bytes are required to encode 1001 bits of data?

Answer: $\lceil 1001/8 \rceil$ = 126 bytes

REFERENCES

- *Discrete Mathematics and its applications with combinatorics and graph theory (7th edition)* by Kenneth H. Rosen [Indian Adaptation by KAMALA KRITHIVASAN], published by McGraw-Hill, is the required text.
- *SCHAUM'S ouTlines Discrete Mathematics(2nd edition)*, by Seymour Lipschutz, Marc Lipson
- *Discrete and Combinatorial Mathematics(3rd Edition)* by *R. P. Grimaldi,*, published by Addison-Wesley